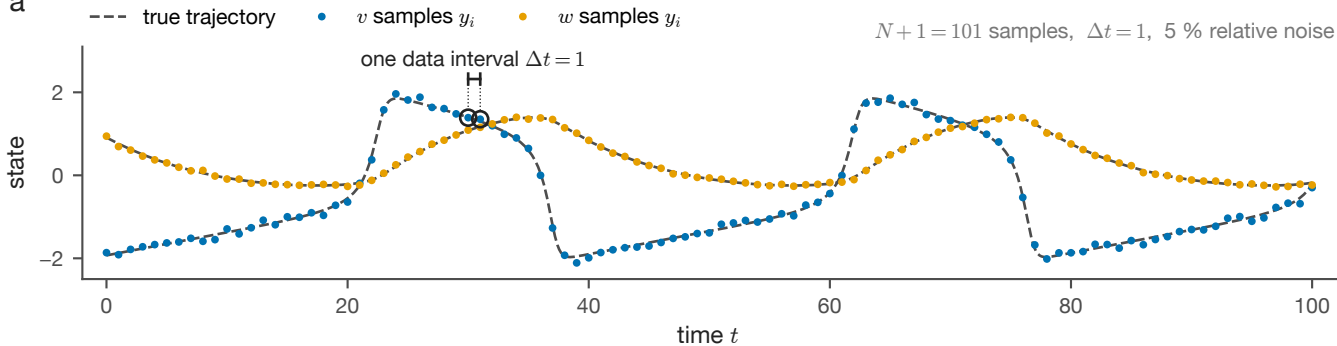
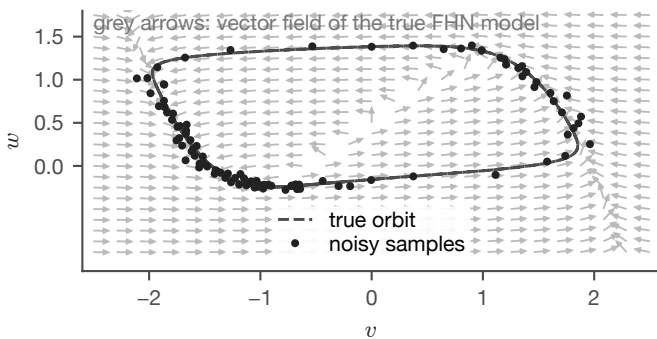


The inverse problem: recover the few non-zero entries of p from noisy samples of v, w

a



b



Data-generating model (FitzHugh–Nagumo)

$$\dot{v} = v - v^3/3 - w + 0.5$$

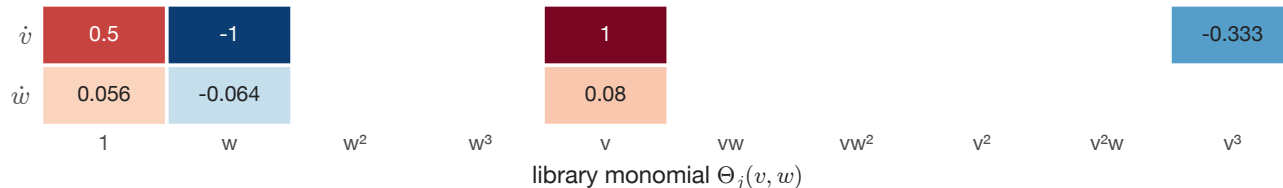
$$\dot{w} = (v + 0.7 - 0.8w) / 12.5$$

Ansatz $\dot{x}_i = \sum_j p_{ij} \Theta_j(v, w)$, where

Θ = every monomial in (v, w) of degree ≤ 3
(10 terms per equation $\Rightarrow$ 20 unknowns p_{ij}).

c

unknown sparse coefficients p (20 entries, 7 non-zero)



blank cell = true zero; cell colour = sign of p_{ij} , shaded by $\sqrt{|p_{ij}|}$